

Turn Data
into Insights 

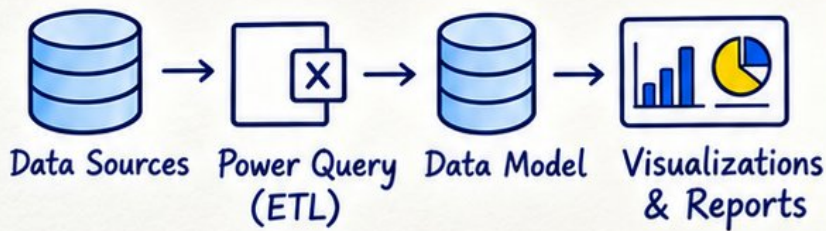


Power BI

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Complete Notes

- ✓ Data Import
- ✓ Data Modeling
- ✓ DAX (Measures)
- ✓ Power Query
- ✓ Visualizations
- ✓ Reports & Dashboards
- ✓ Real Projects



Better Data
Better Decisions



Analyze
Visualize
Share



INTRODUCTION TO POWER BI

1. What is Power BI ?

- Power BI is a Business Intelligence (BI) tool developed by Microsoft. It helps us to connect, transform, visualize and share data to make better business decisions.

2. Business Intelligence (BI) Basics

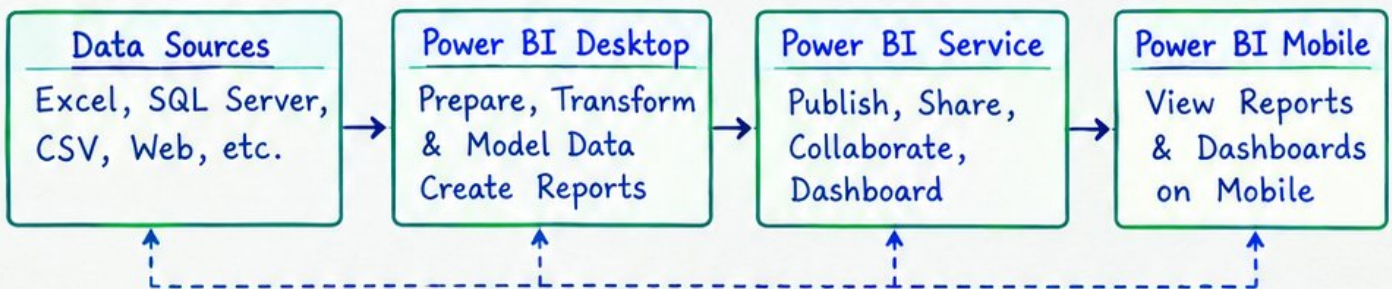
- BI means turning raw data into useful information.

It helps in:

- Data Collection
- Data Analysis
- Data Visualization
- Reporting
- Better Decision Making



3. Power BI Architecture



4. Power BI Components

- Power BI Desktop :**
It is a free Windows application used to connect, transform, model data and create reports.
- Power BI Service :**
It is an online SaaS (Software as a Service) platform. We can publish reports, create dashboards, share and collaborate here.
- Power BI Mobile :**
It is a mobile application (Android, iOS, Windows) to view reports and dashboards on the go.

Note :



5. Power BI Licensing

- Power BI Free :** Free to use Power BI Desktop and basic sharing.
- Power BI Pro :** For sharing, collaboration and larger datasets.
- Power BI Premium :** For advanced features, large organizations and dedicated capacity.

POWER BI INTERFACE

Power BI Desktop is the main tool used to connect data, transform, model and create reports.

Let's understand the Power BI Desktop interface.

The screenshot shows the Power BI Desktop interface with the following components and annotations:

- Ribbon:** Contains all the commands. It includes tabs like File, Home, Insert, Modeling, View, and Help. The Home tab is active, showing groups like Clipboard, Data, Queries, Visuals, Calculations, and Share.
- Views:** (Report, Data, Model) Use these icons to switch between views. The icons are located on the left side of the interface.
- Report Canvas:** Design and build your reports here. It is the central area where reports are created.
- Visualizations Pane:** Choose charts, set values, filters and format here. It contains various chart types and options for values, drill-through, and filters.
- Fields Pane:** Shows all tables and columns from your data. It lists fields like Date, Product, Category, Sub-category, City, Sales, Quantity, and Profit.
- Page Tabs:** Manage pages in your report. It is located at the bottom of the interface, showing 'Page 1' and a '+' button to add new pages.

1. Different Views in Power BI Desktop

- Report View** : Used to create interactive reports using visuals.
- Data View** : Used to preview and explore the data in table format.
- Model View** : Used to create relationships between tables and manage the data model.

Tip :

You can switch between views using the icons on the left side.

2. Important Parts

- Ribbon** : Contains all important commands and tools.
- Views** : Helps to switch between Report, Data and Model views.
- Report Canvas** : Area where we design our reports.
- Visualizations Pane** : Used to select chart type and assign fields.
- Fields Pane** : Shows all tables and columns from the data.
- Page Tabs** : Used to add, delete and manage report pages.

Key Takeaway

Power BI Desktop interface is simple and user friendly. Understanding each part is the first step to becoming a Power BI expert. ★

DATA SOURCES & DATA IMPORT

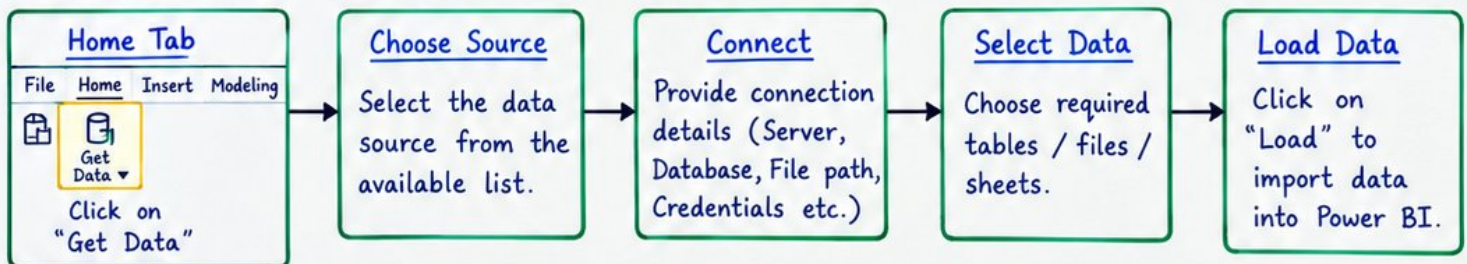
Power BI can connect to a wide variety of data sources, We can import data from these sources and use it to create reports and dashboards.

1. Common Data Sources

- Excel Workbook (.xlsx, .xls)
- CSV Files (.csv)
- Text / Delimited Files
- SQL Server Database
- MySQL Database
- Web Data (Web Page)
- SharePoint Folder / List
- Google Sheets
- Folder (Multiple Files)
- API (Web API)
- Oracle Database
- Azure Services
- Other Databases and Services



2. How to Import Data



3. Data Import Modes

- **Import Mode** : Data is imported and stored in Power BI. Supports fast performance and all Power BI features.
- **DirectQuery Mode** : Data is not imported. Power BI sends queries in real-time to the source. Always up-to-date data but slower.
- **Live Connection** : Connects to Analysis Services / Power BI dataset in real-time. You cannot create calculated columns or tables.

Tip :

For best performance, use Import Mode, Use DirectQuery only for large real-time data.

4. Important Points

- ★ Always clean and prepare data before using in reports.
- ★ Power BI supports scheduled refresh for imported data (in Power BI Service).
- ★ You can combine data from multiple sources.
- ★ Use appropriate data type while importing for better analysis.

Key Takeaway

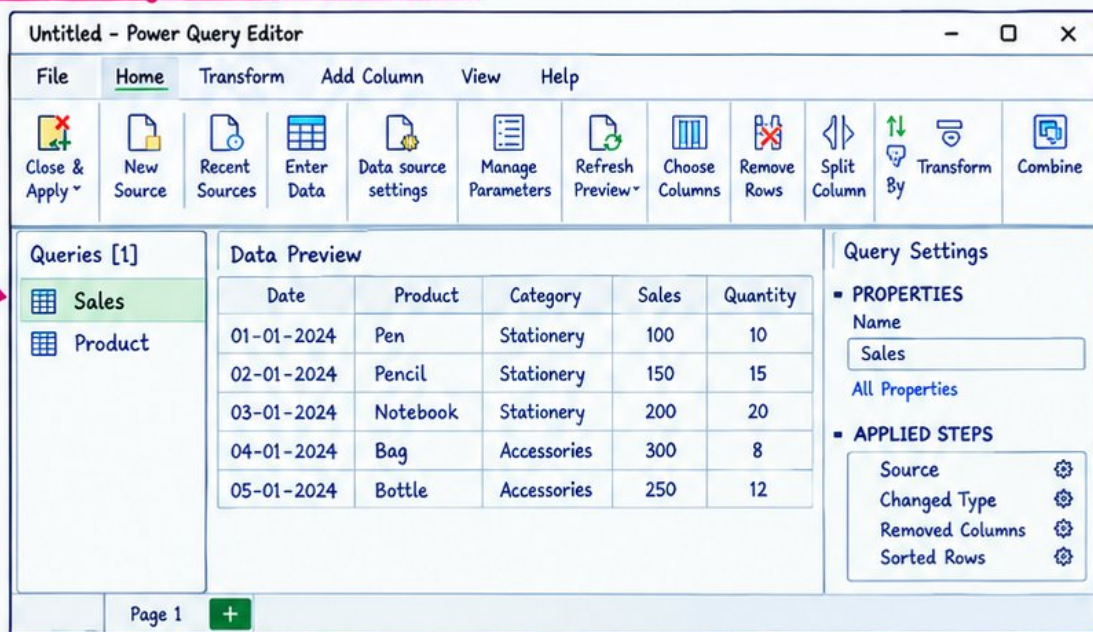
Power BI can connect to many sources and bring data together in one place for analysis and visualization.



POWER QUERY EDITOR (ETL)

Power Query is a data preparation and transformation tool in Power BI. We use it to clean, shape and transform raw data before loading it into the data model.

1. Power Query Editor Interface



Ribbon

Contains all the commands.

Data Preview

Shows the data after each step.

Query Settings

Shows properties and all applied steps.

Queries Pane

Shows all queries in your file.

Page Tabs

Manage pages in your report.

2. Common Data Transformation Operations



Remove Columns - Remove unnecessary columns from data.



Remove Rows - Remove unwanted rows (blank, errors, duplicates etc.).



Split Columns - Split a column into multiple columns.



Merge Columns - Combine two or more columns into one.



Change Data Type - Change the data type (Text, Number, Date etc.).



Replace Values - Find and replace specific values.



Remove Duplicates - Remove duplicate rows.



Append Queries - Add rows from one table to another.



Merge Queries - Combine two tables based on a common column.



Pivot Columns - Turn unique values of a column into new columns.



Unpivot Columns - Convert columns into attribute-value pairs.



Custom Column - Create a new column using custom formula.

Key Takeaway

Power Query helps us to transform raw data into clean, consistent and analysis ready data.

3. Applied Steps

- Every transformation you apply is recorded in the Applied Steps list.
- Steps are executed in order from top to bottom.
- You can add any step by clicking on the gear icon.
- You can delete, reorder or disable steps.
- Steps can be reused for consistency and automation.

Tip:

Always keep your steps simple, reusable and well organized for better performance and easy maintenance.

DATA MODELING IN POWER BI

Data modeling is the process of creating relationships between tables in Power BI. It helps to organize data and make it easier to analyse.

1. What is Data Model?

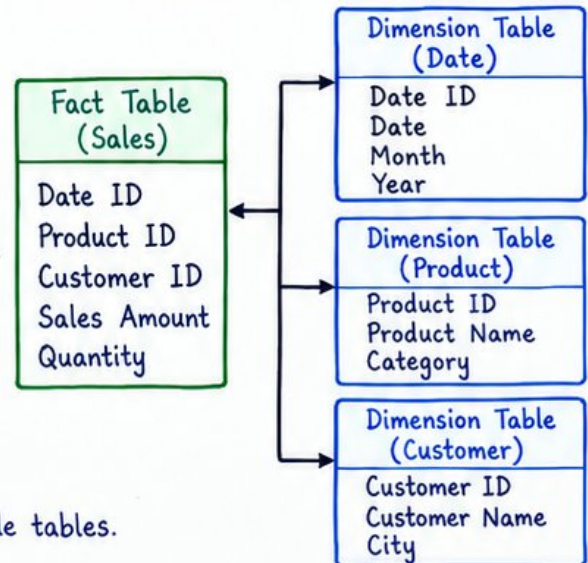
- A data model is a collection of tables, relationships, and calculations (measures and columns).
- It defines how data is organized and how tables are related to each other.

Key Terminology

Good data model = Accurate reports + Better performance + Easy maintenance. ★

2. Types of Tables

- Fact Table** : Contains measurable data (measures).
Example: Sales, Orders, Transactions.
- Dimension Table** : Contains descriptive data (attributes).
Example: Customer, Product, Date, Region.



3. Relationships

- A relationship connects columns from two tables.
- It allows Power BI to combine data from multiple tables.

Types of Relationships:

Type	Diagram	Description	Example
One-to-One (1:1)		One record in table A relates to one record in table B.	Employee Table and Employee Details Table (1:1) relation - 1 record)
One-to-Many (1:*)		One record in table A relates to many records in table B.	Customer (1) → Sales (Many)
Many-to-Many (*:*)		Many records in table A relate to many records in table B.	Student and Course (Students take many courses and each course has many students)

Cardinality

It shows the number of records in one table that relate to records in another table.

1 = One
* = Many

4. Cross Filter Direction

- Single Direction (Default)**



Filters flow from Table A to Table B only.

- Both Direction**



Filters flow in both directions.

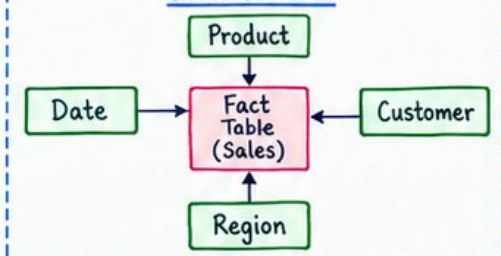
Tip

Use single direction for better performance when you have a specific requirement. 😊

5. Best Practices for Data Modeling

- Use a Star Schema for better performance and easy reporting.
- Keep fact tables in the centre and dimension tables around it.
- Use meaningful and consistent names for tables and columns.
- Create relationships using unique key columns.
- Avoid many-to-many relationships if possible.
- Hide unnecessary columns from report view.

Star Schema



DAX (DATA ANALYSIS EXPRESSIONS)

DAX (Data Analysis Expressions) is a formula language used in Power BI to create calculations in tables or models. DAX is used to create Measures, Calculated Columns and Calculated Tables.

1. Types of DAX Calculations

Type	Description	Created In
Measures	A measure is a calculation that is evaluated on the fly based on the filters in the report.	Created in <u>MEASURES</u> (Created in Values area of visuals)
Calculated Columns	A calculated column adds new data to a table. It is stored in the model.	Created in <u>COLUMNS</u> (Used like any other column)
Calculated Tables	A calculated table creates a new table using a DAX expression.	Created in <u>TABLES</u> (like like a normal table in the model)

Key Terminology

Measures = Dynamic
(changes with filters)

Calculated Column = Static
(calculated in model)

Calculated Table = New table
created using DAX



2. DAX Syntax

Basic Syntax

<Measure> = <Expression>

Example :

Total Sales = SUM(Sales[Amount])

3. Examples of DAX Calculations

- Measures Example :

Total Sales = SUM(Sales[Amount])

→ This measure calculates the total of Amount column in Sales table.

- Calculated Column Example :

Discount Amount = Sales[Amount] * 0.10

→ This column calculates 10% discount for each row.

- Calculated Table Example :

Top Customers = TOPN(10, Customers, Customers[Sales], DESC)

→ This table shows top 10 customers by sales.

Operators in DAX

- Addition
- Subtraction
- Multiplication
- Division
- Power
- & Concatenation

4. Important DAX Elements

- Table Name : Use TableName[ColumnName]
- Functions : DAX has many built-in functions (SUM, COUNT, IF etc.)
Measures use aggregation functions.
- Context : DAX works based on Row Context and Filter Context.

5. Common Aggregate Functions in DAX

Function	Description	Example
SUM()	Adds all values in a column.	Total Sales = SUM(Sales[Amount])
AVERAGE()	Returns the average of values.	Avg Sales = AVERAGE(Sales[Amount])
COUNT()	Counts the number of rows.	Total Orders = COUNT(Sales[Order ID])
DISTINCTCOUNT()	Counts the number of unique values.	Unique Customers = DISTINCTCOUNT(Customers[CustomersID])
MAX()	Returns the maximum value.	Max Sales = MAX(Sales[Amount])
MIN()	Returns the minimum value.	Min Sales = MIN(Sales[Amount])

Note :

DAX is not case sensitive.

Example : sum(), Sum(), SUM() all are valid.

But use proper column names/spelling.



Visualizations help us to represent data in a visual format. Power BI provides a wide range of visuals to analyse and share insights.

1. Commonly Used Visuals in Power BI

Visual	Icon	Description	Best Used For
Table		Displays data in rows and columns.	Detailed data view and quick lookup.
Matrix		Displays data in a hierarchical grid.	Summarized data with subtotals and hierarchies.
Card		Shows a single value.	KPIs, labels, key numbers.
Multi-row Card		Displays multiple values in a card format.	Multiple KPIs in a compact view.
KPI		Shows performance against a goal.	Tracking targets and goals.
Bar Chart		Compares values across categories.	Category comparison (horizontal).
Column Chart		Compares values across categories.	Category comparison (vertical).
Line Chart		Shows trends over a period of time.	Time series data, trends and patterns.
Area Chart		Shows trends with filled area.	Volume over time, accumulative trends.
Pie Chart		Shows parts of a whole.	Percentage distribution (also categories).
Donut Chart		Similar to pie chart with a hole.	Percentage distribution (with better focus).
Scatter Chart		Shows relationship between two values.	Correlation, pattern and outliers.
Map		Shows data on a geographical map.	Geographical analysis by country, state, city.
Treemap		Shows hierarchical data in rectangles.	Proportional comparison & hierarchy.
Gauge		Shows value within a range.	Progress towards a specific target.
Waterfall Chart		Shows cumulative effect of values.	Understand how positive and negative values affect the total.
Funnel Chart		Shows stages in a process.	Process flow, conversion analysis.

Tips :

- Choose the right visual for tell the right story.
- Avoid using too many visuals on a single page.
- Keep your visuals simple, clear and focused.
- Use colors and formatting to highlight important insights.



Which Visuals to Use?

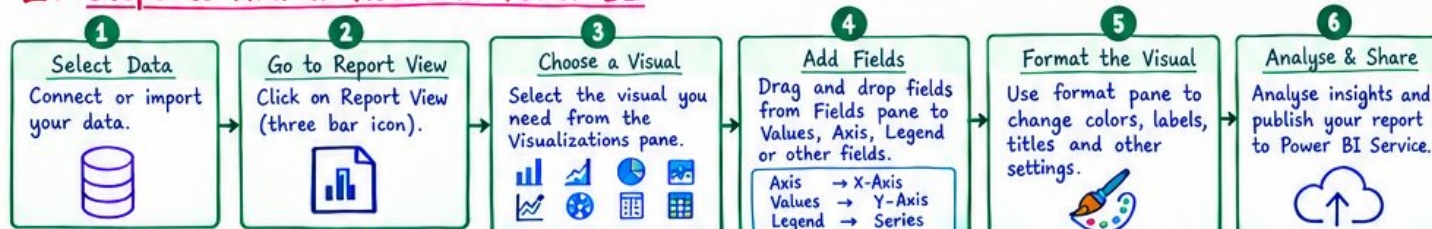
Compare Categories → Bar/Column Chart
 Show Trend Over Time → Line/Area Chart
 Show Part of Whole → Pie/Donut Chart
 Show Relationships → Scatter Chart
 Show Geographical Data → Map
 Show Hierarchical Data → Treemap/Matrix
 Show Progress to Goal → KPI/Gauge
 Show Process/Conversion → Waterfall Chart

Key Takeaway

Visualizations turn your data into meaningful insights. The right visual helps you and your audience understand data quickly and make better decisions.



2. Steps to Add a Visual in Power BI



Note : Always design your report keeping the end user in mind. Make it interactive, simple and informative.

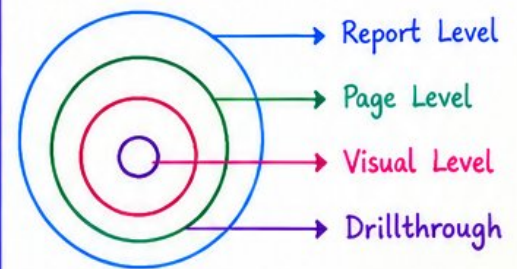


Filters and Slicers help us to focus on specific data in a report. They make reports interactive and easy to use.

1. Types of Filters in Power BI

1. Visual Level Filter : Filters data for a specific visual only.
2. Page Level Filter : Filters data for the entire page.
3. Report Level Filter : Filters data for all pages in the report.
4. Drillthrough Filter : Passes data to a drillthrough page for more detailed analysis.

Filter Levels (Scope)



Note : Filters reduce the data shown in visuals based on the conditions you apply.

2. Slicers in Power BI

- A slicer is a visual filter.
- Users can select values from slicers to filter data interactively.
- Changes made in a slicer automatically update all connected visuals.

Common Slicer Types

- Dropdown Slicer : Shows values in a dropdown list.
- List Slicer : Shows values in a list with checkboxes.
- Date Slicer : Used to filter dates (Year, Quarter, Month, Day).
- Between Slicer : Used to select a range of numbers or dates.

3. Difference Between Filters and Slicers

Feature	Filters	Slicers
Purpose	Restrict data shown in visuals.	Allow user to interact and filter data.
Visibility	Usually hidden (in filter pane).	Visible on the report page.
User Interaction	Not directly visible to users.	Users can click and select values.
Types	Visual, Page, Report, Drillthrough.	Dropdown, List, Date, Between.
Best Used For	Page/Report level filtering or advanced conditions.	Quick filtering and interactive analysis.

Tips

- ★ Use slicers for better user experience.
- ★ Sync slicers across pages for consistency.
- ★ Use "Edit Interactions" to control which visuals respond to slicers.

4. Filter Pane in Power BI



→ Filters on this visual : Applies only to selected visual.

→ Filters on this page : Applies to all visuals on the current page.

→ Filters on all pages : Applies to all pages in the report.

5. How to Add a Slicer

- 1 Go to Report View (📊).
- 2 Click on Slicer icon (in Visualizations pane).
- 3 Drag and drop the field into the slicer field well.
- 4 Choose slicer type from the dropdown (if enabled).
- 5 Use Format pane to change style, title, etc.
- 6 Set the slicer by selecting values.

Slicer Example (Region) ✕

- ☐ Select all
- ☒ North
- ☐ South
- ☐ East
- ☐ West

Key Takeaway

Filters and Slicers are powerful tools in Power BI that help users analyse the right data, quickly. Use Filters for setting rules and slicers for user-friendly interaction.



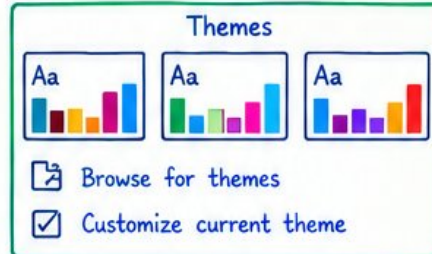
Good formatting and design make your report attractive, easy to read and more effective in delivering insights.

1. Themes in Power BI

- Themes control the overall look and feel of your report.
- You can apply a built-in theme or create a custom theme.
- Themes change colors, fonts and effects across the report.

Steps to Apply a Theme :

- Go to "View" tab.
- Click on "Themes".
- Choose a theme from the list.
- Your report design will change instantly.



Best Practices

- ★ Use consistent colors throughout the report.
- ★ Keep background simple and clean.
- ★ Use readability-friendly fonts and sizes.
- ★ Highlight important numbers and KPIs.



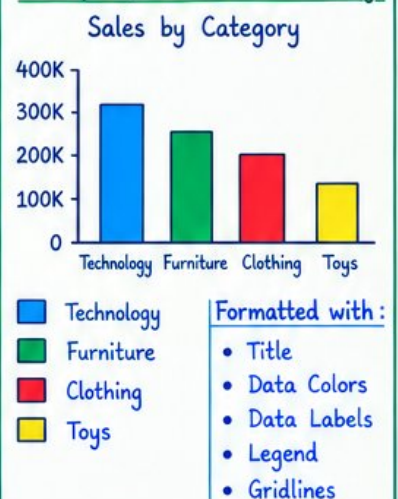
2. Formatting Visuals

You can format each visual to improve clarity and highlight key information.

Common Formatting Options :

Option	Description	Examples / Use
Title	Add or change the title of the visual.	Add meaningful title to charts.
Background	Change background color or add transparency.	Use light colors for better look.
Data Colors	Change colors of data points.	Use different colors for categories.
Data Labels	Show values on charts.	Show value on bars or pie slices.
Axis	Format X-axis and Y-axis.	Change font, color, title, position.
Legend	Show or hide legend and format it.	Change position and font size.
Gridlines	Show or hide gridlines.	Helps in better readability.
Tooltips	Customize tooltip information.	Add more details in tooltips.
Header & Footer	Add header or footer to visual.	Shows visual short info.

Example (Bar Chart Formatting)



3. Conditional Formatting

- Conditional formatting helps to highlight important data based on conditions.
- It quickly draws attention to key trends and exceptions.

Common Uses :

- Highlight high and low values.
- Show data bars, color scales, icons.
- Make your data easy to understand.

Sales Dashboard

Performance
(Revenue vs Target)

Steps to Conditional Formatting :

- Data Bars - Adds bars to cells based on value.
- Color Scales - Applies color background to cells.
- Icon Sets - Displays icons based on values.

Steps to Apply :

- Select the field or measure.
- Go to "Format" pane.
- Click on conditional formatting (fx).
- Choose the rule and set the conditions.
- Click OK and apply.

Tip :

Use conditional formatting for KPIs, top/low performers, trends and exceptions.



4. Additional Design Elements

Title & Text

- Use clear and meaningful titles.
- Add descriptions or notes if needed.

Sales Dashboard

Revenue vs Target
Performance

Background

- Use light and simple backgrounds.
- Avoid heavy patterns.



Shapes & Icons

- Use shapes, icons and images to represent understanding.
- Keep it simple.



Alignment

- Align visuals properly.
- Keep equal spacing between elements.



Font & Size

- Use readable fonts.
- Use proper text size for titles, labels and values.

Aa Aa Aa

Key Takeaway

A well formatted and designed report improves readability, grabs attention and helps users understand the insights quickly and make better decisions.



Power BI can connect to many data sources. The "Get Data" feature helps us to import data into Power BI for analysis and reporting.

1. What is a Data Source?

- A data source is the place where data is stored.
- Power BI connects to the data source and brings the data into Power BI.

2. Types of Data Sources in Power BI

Category	Examples	Description
File	• Excel Workbooks (.xlsx) • CSV, Text Files • XML, JSON Files	Data stored in files on your computer or network.
Database	• SQL Server • MySQL, PostgreSQL • Oracle, IBM DB2	Data stored in relational databases.
Cloud	• Azure SQL Database • Azure Blob Storage • Google BigQuery	Data stored in cloud services.
Online Services	• SharePoint List • Salesforce • Google Analytics	Data available from online platforms.
Other	• Web (HTML) • OData Feed • Blank Query	Web pages, APIs, or other advanced sources.

Key Takeaway

Power BI supports many data sources. Choose the right source based on your data location and business need. ★

3. Get Data in Power BI

The "Get Data" feature is used to connect and import data into Power BI.

Steps to Get Data :

- ① Open Power BI Desktop.
- ② Go to Home tab.
- ③ Click on Get Data 
- ④ Choose the data source.
- ⑤ Provide connection details if required.
- ⑥ Select the tables or files you want.
- ⑦ Click Load to import data or Transform Data to clean data first.

4. Get Data Flow in Power BI



Tips :

- ★ Manage preview data before loading.
- ★ Use Transform Data to clean and change your data.
- ★ Import only the data you need.
- ★ Use appropriate data types for better analysis. 😊

5. Data Import Modes

Mode	Description	Best Used For
Import	Data is imported and stored in Power BI model.	Faster performance and best for small to medium data.
DirectQuery	Data is not stored in Power BI. It queries the source in real time.	Large datasets and live data from databases.
Dual	Combination of Import and DirectQuery.	Large models with aggregations.

Import vs DirectQuery

Import

- Faster performance
- Data stored in model
- Works offline
- Needs data refresh

DirectQuery

- Real-time data
- Data not stored
- Always connected
- Slower performance

6. Import/Direct Query in Practice

- Choose your data source and provide it to the data source.
- Use filtering and source connection.
- Refresh data regularly to keep reports up to date.
- Use appropriate mode (Import/DirectQuery) based on data size and need.
- Clean and transform data for accurate analysis.

Common Data Sources Handled

 Excel : Excel files  SQL Server : Connect to SQL databases  Web : Import data from web pages	 SharePoint : Get data from SharePoint lists  Salesforce : Import data from CRM  Blank Query : Create custom queries
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Key Takeaway

Choosing the right data source and using Get Data correctly is the first step towards building powerful and accurate reports in Power BI. ★

Data modeling organizes data, creates relationships and ensures accurate analysis in Power BI.

1. What is Data Modeling?

- It is the process of organizing data from different sources.
- It involves tables, relationships and calculations.
- It helps in fast queries and accurate results.

2. Components of Data Model

Component	Description	Purpose	Icon
Tables	Collection of related data in rows and columns.	Stores and organizes data.	
Relationships	Connection between tables using matching columns.	Combines data from multiple tables.	
Columns	Fields in a table.	Contains data items.	
Measures	Data formulas that return a single value.	Performs calculations on data.	
Calculated Columns	New columns created using DAX for each row.	Add new information to the table.	
Hierarchies	Organizes data in levels.	Enables drill down analysis.	

Key Takeaway

A well-designed data model makes reports faster, easier to understand and more reliable.



3. Types of Relationships

1. One-to-One (1:1)



- Each row in Table A relates to one row in Table B.

2. One-to-Many (1:M)



- One row in Table A relates to many rows in Table B.

3. Many-to-Many (M:M)



- Rows in Table A relate to many rows in Table B and vice versa.

4. Cardinality Relationships

Steps :

- Go to Model view.
- Drag the column from one table to another.
- Power BI automatically detects the relationship.
- Set Cardinality and Cross Filter direction.
- Click OK.

Example: One-to-Many Relationship

Customer Table

CustomerID	Name
1	John
2	Neha
3	Raj

Sales Table

SalesID	CustomerID	Amount
101	1	1000
102	1	2000
103	2	1500
104	3	2500

One customer can have many sales, but each sale belongs to one customer.

5. Cross Filter Direction

Type	Description	Diagram
Single	Filters flow in one direction (default).	
Both	Filters flow in both directions.	
None	No filter is passed between tables.	

6. Best Practices for Data Modeling

- Use meaningful table and column names.
- Remove unnecessary columns and duplicates.
- Create proper relationships between tables.
- Use a Date Table for time intelligence.
- Keep fact tables in the center.
- Use dimensions in descriptive tables.
- Avoid bi-directional relationships where needed.
- Test the model for performance.

Star Schema (Recommended)



- Fact table contains measures.
- Dimension tables describe the facts.
- Improves performance and simplifies reports.

7. Role of Data Table

Benefit	Description
Time Intelligence	Enables advanced calculations like YTD, MTD, QTD.
Conditional Analysis	Performs calculations based on conditions in data.
Better Filtering	Helps in easy filtering by year, month, quarter, etc.

8. Common DAX in Data Model

DAX Function	Syntax Example	Use
SUM	Total Sales = SUM(Sales[Amount])	Adds values in a column.
COUNTROWS	Total Orders = COUNTROWS(Orders)	Counts number of rows.
CALCULATE	Total Revenue = CALCULATE(SUM(Sales[Amount]), FILTER(Region, Region[Region] = "East"))	Evaluates in a modified filter context.
RELATED	City = RELATED(Customer[City])	Gets data from related table.
ALL	All Sales = CALCULATE(SUM(Sales[Amount]), ALL(Sales))	Removes filters.



Tip :

Good data modeling is the foundation of powerful and fast dashboards.



A strong data model converts raw data into the right story, to deliver the right insights!



Topic - 12

DAX (DATA ANALYSIS EXPRESSIONS) IN POWER BI

Date : ___/___/___

DAX is the formula language used in Power BI to create calculations, measures, calculated columns and calculated tables.

Key Takeaway

DAX helps you perform powerful calculations and create custom insights in Power BI.



1. Where Can We Use DAX?

Object	Description	Example
Measures	Dynamic calculations that change with filters.	Total Sales = SUM(Sales[Amount])
Calculated Columns	New column added to a table, calculated row by row.	Discounted Price = Sales[Price] * 0.9
Calculated Tables	New tables created using DAX expressions.	Top 10 Products = TOPN(10, Products, Sales[Amount], DESC)

2. Basic DAX Syntax

Result = FUNCTION(Column1, Column2, ...,)

↓ ↓ ↓

Name of the measure or calculated column DAX function Columns, values or expressions

3. Commonly Used DAX Functions

Function	Syntax	Description	Example
SUM	SUM([Column])	Adds all numbers in a column.	SUM(Sales[Amount])
AVERAGE	AVERAGE([Column])	Returns the average of numbers.	AVERAGE(Sales[Amount])
COUNTROWS	COUNTROWS(Table)	Counts the number of rows in a table.	COUNTROWS(Sales)
CALCULATE	CALCULATE(Expression, Filter)	Modifies filter context and evaluates an expression.	CALCULATE(SUM(Sales[Amount]), Sales[Year] = 2024)
FILTER	FILTER(Table, Condition)	Returns a table that satisfies the condition.	FILTER(Sales, Sales[Amount] > 100)
ALL	ALL(Table/Column)	Removes all filters from the table/column.	ALL(Sales)
RELATED	RELATED(Column)	Gets related value from another table.	RELATED(Customer[Country])

Real Functions

- ★ The measure instead of calculated columns for better performance.
- ★ Use meaningful names for your DAX objects.
- ★ Test your expressions by using Dax.
- ★ Understand filter context in CALCULATE.



4. Measures Examples

Measure	DAX Expression
Total Sales	Total Sales = SUM(Sales[Amount])
Total Quantity	Total Quantity = SUM(Sales[Quantity])
Average Price	Average Price = AVERAGE(Sales[Price])
Total Orders	Total Orders = COUNTROWS(Sales)
Sales in 2024	Sales in 2024 = CALCULATE(SUM(Sales[Amount]), Sales[Year] = 2024)
Sales to Date	Sales to Date = TOTALYTD(SUM(Sales[Amount]), Sales[Date])
Total Customers	Total Customers = DISTINCTCOUNT(Sales[CustomerID])

5. Calculated Column Example

In Sales Table

Discounted Price = Sales[Price] * 0.9

Profit = Sales[Amount] - Sales[Cost]

Year = YEAR(Sales[OrderDate])

Calculated columns increase the size of your data model.

6. Calculated Table Example

New Table: High Value Sales

High Value Sales =

FILTER(Sales, Sales[Amount] > 1000)

Calculated tables are static and created during data refresh.

7. Filter Context vs Row Context

Filter Context	Row Context
- Determines which rows are available in a calculation. - Comes from slicers, filters, visuals or CALCULATE. - Example: Total Sales = SUM(Sales[Amount])	- Deals with a formula in individual row by row. - Created from calculated columns or iterators. - Example: Discounted Price = Sales[Price] * 0.9

VS

8. CALCULATE Function in Action

Original Measure

Total Sales = SUM(Sales[Amount])



With CALCULATE

Sales 2024 = CALCULATE(SUM(Sales[Amount]), Sales[Year] = 2024)

CALCULATE changes the filter context, so the measure returns value for Year = 2024 only.

9. Time Intelligence Functions (Examples)

Function	Syntax	Use
TOTALYTD	TOTALYTD(Expression, DateColumn)	Year to date total
SAMEPERIODLASTYEAR	SAMEPERIODLASTYEAR(DateColumn)	Same period last year
DATEADD	DATEADD(DateColumn, -1, YEAR)	Move date by a period

Example

Total Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))

10. DAX Tips

- ✓ Use variables (VAR) to simplify complex expressions.
- ✓ Use DIVIDE() instead of / to avoid divide by zero errors.
- ✓ Use FORMAT() for formatting numbers or dates.
- ✓ Optimize your DAX for performance.



DAX is the heart of Power BI analysis. Master DAX to create powerful calculations, deeper insights and advanced reports that drive better decisions.



Topic - 13

NAVIGATION & INTERACTIVITY IN POWER BI

Date : __/__/__

Key Takeaway

- ★ Good navigation and interactivity help users explore data easily and get better insights. 😊

Navigation and interactivity make Power BI reports user-friendly, dynamic and easy to explore.

1. Why Navigation & Interactivity?

- Help users move between pages and visuals.
- Allow filtering and highlighting of data.
- Improve user experience and engagement.
- Make reports more dynamic and insightful.



2. Types of Navigation

Type	Description	Icon	Example
Page Navigation	Move between pages in a report.		Buttons, Bookmarks
Bookmark Navigation	Save and go to a specific view.		Bookmarks
Drill Navigation	View details in more detail.		Drill down, Drill through
Custom Navigation	Use shapes, images or buttons.		Image maps, Menus

3. Navigation Using Buttons

Button	Action	Icon
Page Navigator	Go to another page.	➔
Bookmark Navigator	Go to a saved bookmark.	
Back	Go back to previous page.	⬅️
Info	Show additional information.	
Clear Filters	Reset all applied filters.	

Examples

Home

Next Page ➔

⬅️ Back

Info

Clear Filters

You can find buttons in Insert tab under Buttons.

4. Bookmarks in Power BI

Steps to Create a Bookmark:

- 1 Apply filters, slicers or other changes to your report.
- 2 Go to View tab → Bookmarks pane.
- 3 Click Add to create a bookmark.
- 4 Give it a name.
- 5 Use a button to navigate to the bookmark.

Bookmark Options

- Data (default)
- Display
- Current Page
- All Visuals



Bookmarks help save multiple states of the report and switch between them.

5. Drill Down, Drill Up & Drill Through

Drill Down	Drill Up	Drill Through
Move from a higher level to detail.	Go back to a higher level of data.	Navigate to a different page with more details.
Category → Subcategory	Subcategory → Category	From summary to detailed page.

Steps to Enable Drill Through

- 1 Create a new detailed page.
- 2 Add fields that will act as Drill Through Fields.
- 3 Go to Visuals pane and select the fields.
- 4 In Properties pane → Turn on Drill Through.
- 5 Now, users can right-click a data point and select Drill Through.

Properties

General

Action

Drill through

Drill through

6. Slicers for Interactivity

Slicer Types

- ☐ List
- ☐ Dropdown
- ☒ Tiles
- ☐ Between
- ☐ Relative

Slicer Tips

- ✓ Use slicers to filter visuals on the page.
- ✓ Supports single select and multi-select.
- ✓ Use a dropdown list or between option.
- ✓ Sync slicers across pages for consistency.
- ✓ Add clear title for better understanding.



Slicers make reports interactive and user-friendly.

7. Sync Slicers Across Pages

Steps

- 1 Add a Slicer to a page.
- 2 Go to View tab.
- 3 Click Sync slicers.
- 4 Check the pages where you want to sync.
- 5 Click OK.

Sync Slicers Panel

Page Name	Sync
Overview	☒
Sales	☒
Products	☐
Region Analysis	☐

Slicers will now sync on the selected pages.

8. Report Tooltips

Tooltips display additional information when users hover over a visual.

Steps to Enable a Tooltip Page

- 1 Create a new page.
- 2 Turn off Page Navigation.
- 3 Design the tooltip content.
- 4 Go to Page Information → Tooltip (On).
- 5 Hover over the visual to see the tooltip.

Example



Tooltip Best Practices

- ✓ Keep it short and simple.
- ✓ Use clear and readable font.
- ✓ Show meaningful data only.
- ✓ Improve user experience. 💡

9. Other Interactivity Features

Highlights

Highlight data points in visuals.

Cross Filter

Filter visuals across pages.

Q&A Visual

Ask questions in natural language and get answers.

Smart Narratives

Automatically generate text insights for your data.

Personalize Visuals

Allow users to customize their visuals.



Good navigation and interactivity turn reports into powerful storytelling tools, helping users explore data, discover insights and take better decisions.



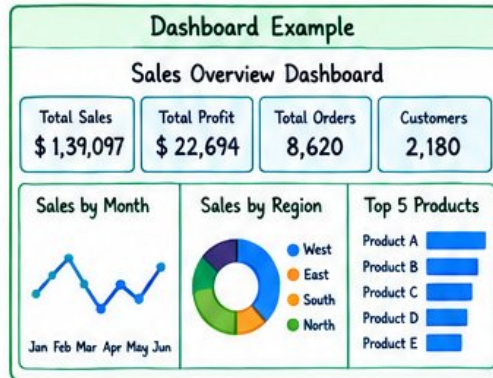
Dashboards give a quick overview of key insights. Publishing makes your reports accessible and shareable in the Power BI Service.

Key Takeaway

- ★ Dashboards provide an at-a-glance view of important data and publishing helps you share insights securely with others. 😊

1. Dashboard in Power BI

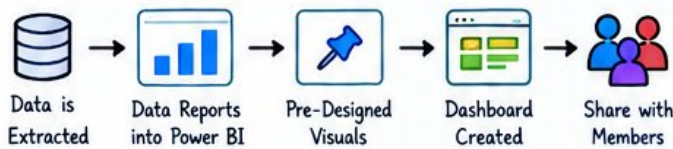
- A dashboard is a single page view of important visuals from one or more reports.
- It is used for monitoring KPIs and business metrics.
- Dashboards are interactive and update with real-time data.



2. Dashboard vs Report

Feature	Dashboard	Report
Purpose	Monitor KPIs and key metrics	Analyze and explore detailed data
Data Scope	Can use multiple reports	Based on a single dataset
Interactivity	High-level interactivity	Detailed drill and explorations
Layout	Single page	Multiple pages
Best For	Quick insights and monitoring	In-depth analysis and storytelling

3. Flow: Dashboards Work



💡 You can pin visuals from multiple reports in a single dashboard.

💡 Keep dashboards simple, focused and updated.
Too many visuals can make it cluttered.

4. Publishing a Report

Steps to Publish:

- 1 Click File → Publish.
- 2 Select a destination (workspace).
- 3 Click Select and then Publish.
- 4 View the report in Power BI Service.
- 5 View the report in Power BI Service.



5. Workspace in Power BI Service

- Workspaces are collections for data, manage and share reports, dashboards and datasets.
- They help in collaboration and access management.

Types of Workspaces	
Type	Description
My Workspace	Personal workspace for individuals
Shared Workspace	Used for teams to collaborate
App Workspace	Used to share with end users

6. Sharing Options

Method	Description	Use Case
Share Workspace	Share a workspace with users or groups.	Team sharing of reports
Share Report	Share a report with specific permissions.	Selected users access
Publish to Web	Embed and share with anyone (public).	For public or external audience
Mobile Access	Access dashboards and reports on mobile devices.	On-the-go insights

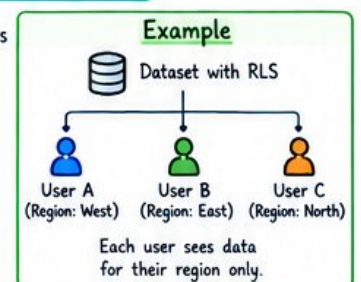
7. Refresh Data in Service

- Keeps your dashboards and reports up to date by refreshing data.
- You can schedule refresh or refresh manually.

Refresh Methods	Refresh Modes	Note:
Scheduled Refresh (set a schedule for auto refresh) Manual Refresh (refresh anytime)	Whole dataset Specific tables Incremental refresh (for large data)	Scheduled refresh requires a data gateway if your data source is on-premises.

8. Row-Level Security (RLS) in Service

- RLS allows you to restrict data access based on user roles.
- Users can only see data they are allowed to see.



9. Dashboard Tiles

- Dashboard is made up of tiles.
- Each tile represents a visual or KPI.
- You can customize tiles, pin them and filtering.

Tile Actions

- ✓ Pin
- ✓ Move
- ✓ Resize
- ✓ Set as default
- ✓ Delete

Tile Types

- Visual Tile (chart, table, etc.)
- KPI Tile
- Map Tile
- Web Content Tile

10. Best Practices

- ✓ Keep dashboards clean & clear purpose.
- ✓ Use only key metrics and insights.
- ✓ Use consistent colors and formatting.
- ✓ Keep dashboards updated and easy to read.
- ✓ Set up scheduled refresh for latest data.
- ✓ Manage access and security properly.
- ✓ Test dashboards on different devices.



A great dashboard tells the story of a glance, and publishing ensures that story reaches the right audience!



Topic-15

POWER BI POWER QUERY EDITOR (DATA TRANSFORMATION)

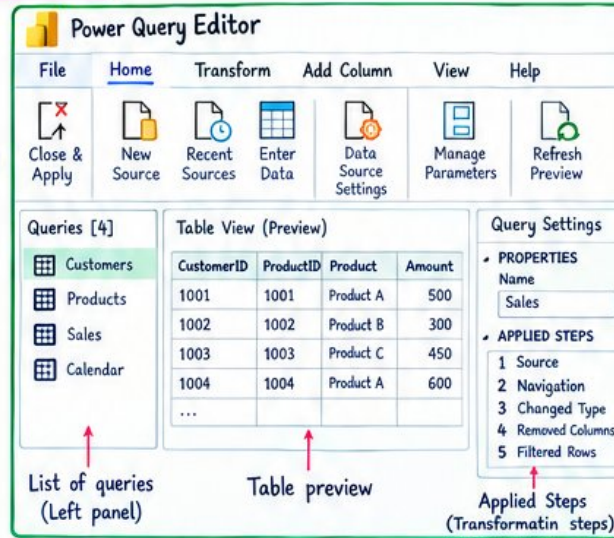
Key Takeaway

- ★ Power Query helps convert raw data into clean, suitable and analysis-ready data. 😊

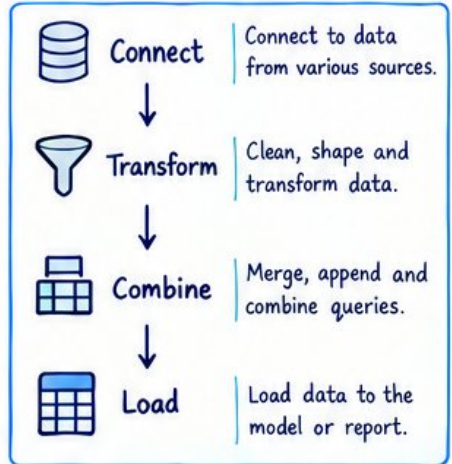
Power Query Editor is the heart of data transformation in Power BI. It helps you clean, shape and prepare data before loading it to the model.

1. What is Power Query Editor?

- It is a data connection and transformation tool.
- You can clean, combine and reshape data without writing any code.
- All steps are recorded in the Applied Steps pane.



2. Power Query Workflow



3. Common Data Sources

Category	Examples	Icon
Files	Excel, CSV, Text/CSV, JSON, XML, Folder	
Databases	SQL Server, Azure, Oracle, MySQL, PostgreSQL	
Online Services	SharePoint, OneDrive, Web, OData, Salesforce	
Other	Azure, Google, Power BI, Facebook, R, Python	

4. Conversion Transformations

Transformation	Description
Remove Rows	Remove unwanted rows.
Remove Columns	Remove unnecessary columns.
Change Data Type	Change the data type of a column.
Split Column	Split one column into multiple columns.
Merge Columns	Combine multiple columns into one.
Filter Rows	Keep rows that meet a condition.
Text Before/After	Extract text before or after a delimiter.
Replace Values	Find and replace values in a column.
Pivot/Unpivot	Change data from rows to columns or vice versa.

5. Applied Steps

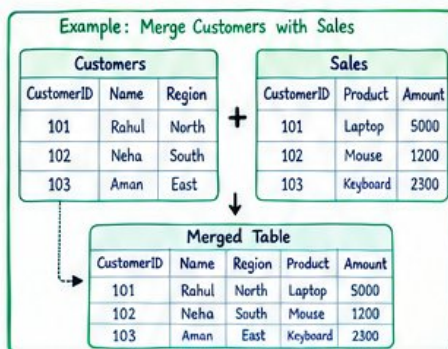
- Every action in Power Query is recorded as a step.
- Steps are applied from top to bottom.
- You can edit, delete, reorder or insert steps.

Tip

- ✓ Rename queries (right click).
- ✓ Remove unnecessary steps.
- ✓ Keep transformation steps few.
- ✓ Use data types correctly.

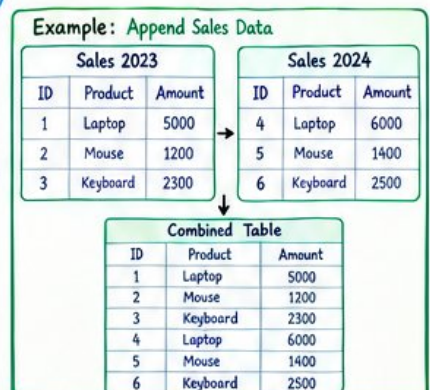
6. Merge Queries

- Merge combines columns from two queries based on a related column.
- It works like a VLOOKUP.



7. Append Queries

- Append adds rows from one or more queries with the same columns.
- It works like a UNION ALL.



8. Useful Merge Tables

- Merge tables helps you combine data from different tables using a common field.
- It is useful for adding details like customer info, product info, etc.

Merge Table	
Field	Example
Customer ID	101, 102, 103
Product ID	P01, P02, P03
Date	01-01-2024
Region	North, South, East

9. Useful Power Table

- Power Table is used to create a structured table view.
- It helps in organizing data with proper formatting.

Table Elements	
	Add Column
	Remove Column
	Sort
	Filter
	Change Data Type

10. Best Practices

- ✓ Keep your data clean and consistent.
- ✓ Use correct data types.
- ✓ Remove duplicates.
- ✓ Use parameters for dynamic data.
- ✓ Document your steps properly.
- ✓ Test the output on different datasets.
- ✓ Maintain source files and backups.



A great dashboard tells the story of a glance,
and powerful data transformation ensures that story is accurate!

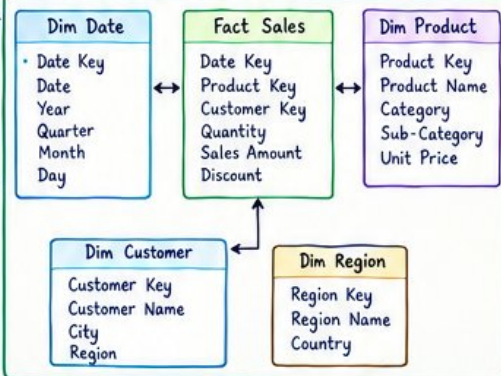


Data modeling organizes data into tables and relationships to create a powerful foundation for reports and analysis.

1. What is Data Modeling?

- Data modeling refers to the logical (tabular) structure of data in Power BI.
- It helps represent real-world business logic.
- A good model is simple, accurate and efficient.

Example: Sales Data Model



2. Types of Relationships

Relationship	Description	Example	Cardinality
One-to-Many	One record in a table related to many records in another table.	Customer → Orders	1 → *
Many-to-One	Many records relate to one record.	Order → Customer	* → 1
One-to-One	A record in a table relates to one record in another table.	User → User Details	1 ↔ 1
Many-to-Many	Many records in a table relate to many records in another table.	Product → Category	* ↔ *

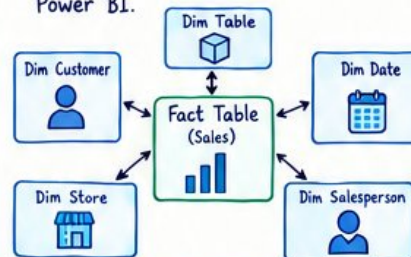
3. Cardinality and Cross Filter Direction

Cardinality	Cross Filter Direction	
1: One unique value in the "one" side.	Single (Default)	Both
*: Many / Multiple values in the "many" side.	1 → *	1 ↔ 1
	Filters flow in one direction.	Filters flow in both directions.

You can use single or both direction based on your business requirement and model design.

4. Star Schema (Recommended)

- A star schema has a central fact table surrounded by dimension tables.
- It is easy to understand, perform well and used in Power BI.



Star schema is the best practice for most Power BI models.

5. Applied Steps

- Every action in Power Query is recorded as a step.
- Steps are applied from top to bottom.
- You can edit, delete, reorder or insert steps.

Tip

- ✓ Rename queries (right click).
- ✓ Remove unnecessary steps.
- ✓ Keep transformation steps few.
- ✓ Use data types correctly.

6. Merge Queries

- Merge combines columns from two queries based on a related column.
- It works like a VLOOKUP.

Customers			Sales		
CustomerID	Name	Region	CustomerID	Product	Amount
101	Rahul	North	101	Laptop	5000
102	Neha	South	102	Mouse	1200
103	Aman	East	103	Keyboard	2300

↓

Merged Table					
CustomerID	Name	Region	Product	Amount	
101	Rahul	North	Laptop	5000	
102	Neha	South	Mouse	1200	
103	Aman	East	Keyboard	2300	

Note:

Scheduled refresh requires a data gateway if your data source is on-premises.

7. Append Queries

- Append adds rows from one or more queries with the same columns.
- It works like a UNION ALL.

Query 1			Query 2		
ID	Name	Sales	ID	Name	Sales
1	A	1000	4	D	4000
2	B	2000	5	E	5000
3	C	3000	6	F	6000

↓

Combined Result		
ID	Name	Sales
1	A	1000
2	B	2000
3	C	3000
4	D	4000
5	E	5000
6	F	6000

8. Useful Merge Tables

- Merge tables helps you combine data from different tables using a common field.
- It is useful for adding details like customer info, product info, etc.

Merge Table				
Customer ID	Name	Product	Region	
101	Rahul	Laptop	North	
102	Neha	Mouse	South	
103	Aman	Keyboard	East	

9. Useful Power Table

- Power Table is used to create a structured table view.
- It helps in organizing data with proper formatting.

Table Elements	
	Add Column
	Remove Column
	Sort
	Filter
	Change Data Type

10. Best Practices

- ✓ Keep your data clean and consistent.
- ✓ Use correct data types.
- ✓ Remove duplicates.
- ✓ Use parameters for dynamic data.
- ✓ Document your steps properly.
- ✓ Test the output on different datasets.
- ✓ Maintain source files and backups.



11. Checklist

Item	Key Actions	Done
Source Data	Verify source & check data quality	☐
Transform	Clean, shape and standardize data	☐
Model	Create relationships and set cardinality	☐
Visualize	Build reports and validate results	☐

12. Helpful DAX Functions for Modeling

Function	Use
RELATED()	Get related values from another table
RELATEDTABLE()	Get all related rows from another table
LOOKUPVALUE()	Return a value from another table
USERELATIONSHIP()	Activate inactive relationship

13. Conclusion

- ✓ A good data model is the foundation of great Power BI reports.
- ✓ Modeling helps in better performance, accuracy and flexibility.
- ✓ Follow best practices and keep your model simple and clean.



≤ A well-structured data model turns your data into meaningful insights! ≤



DAX is the formula language of Power BI. It helps you create calculations, measures, and columns to perform advanced analysis.

1. What is DAX?

- DAX (Data Analysis Expressions) is a formula language.
- It is used to create measures, calculated columns and tables.
- DAX functions along with other BI tools.
- DAX is case-insensitive.

Example DAX Formula

= SUM(Table[Sales Amount])

Function

Column Reference

2. Types of DAX Calculations

Type	Description	Example
Measure	Calculated at query time (does not store data).	Total Sales = SUM(Sales[Amount])
Calculated Column	Creates a new column in the model.	Sales Amount = Sales[Quantity] * Sales[Price]
Calculated Table	Creates a new table using DAX.	Top 5 Customers = TOPN(5, Customer, [Total Sales])

Key Takeaway

- DAX turns your data into meaningful insights with powerful calculations. 😊

Tip

- Use measures for dynamic results.
- Use calculated columns for row-level calculations.
- Use calculated tables for advanced analysis.

3. Basic DAX Syntax

= Function Arguments Expression

Structure:

- Functions perform operations.
- Arguments are the values, column or expression.

4. Common DAX Functions

Category	Function	Description	Example
Aggregation	SUM()	Adds up values.	= SUM(Sales[Amount])
	AVERAGE()	Calculates average.	= AVERAGE(Sales[Amount])
	COUNT()	Counts non-blank values.	= COUNT(Sales[CustomerID])
Text	CONCATENATE()	Joins text values.	= CONCATENATE("Hello ", Customer[Name])
Logical	IF()	Returns value based on condition.	= IF(Sales[Amount] > 100, "High", "Low")
Time Intelligence	TOTALYTD()	Calculates year-to-date totals.	= TOTALYTD([Total Sales], Date[Date])

5. Creating a Measure

- Go to Modeling tab.
- Click New Measure.
- Write the DAX formula.
- Press Enter.
- Use the measure in visuals.

Tip

Use measures for dynamic and flexible calculations.

6. Filter Context in DAX

- Filter context defines the data visible in a DAX expression.
- It can come from slicers, filters, or relationships.
- CALCULATE() can modify filter context.

How Filter Context Works



Filters reduce data; DAX calculates with that filtered data.

7. CALCULATE() Example

Changes filter context to calculate a different result.

Total Sales = CALCULATE(SUM(Sales[Amount]), Sales[Region] = "North")

With CALCULATE():

Total Sales (North) =
CALCULATE(SUM(Sales[Amount]),
Sales[Region] = "North")

Applies new filter context
Changes the result

8. Row Context vs. Filter Context

	Row Context	Filter Context
	Works on individual rows (e.g., in calculated columns).	Works on a set of rows (e.g., in measures).
	Used in calculated columns.	Used in measures and visuals.
	Created by iterators (ADDCOLUMNS, SUMX, etc.).	Created by filters or relationships.
	Fixed for each row.	Can change based on slicers/filters.

9. Useful Power Table

- Power Table is used to create a structured table view.
- It helps in organizing data with proper formatting.

Table Elements

- Add Column
- Remove Column
- Sort
- Filter
- Change Data Type

10. Best Practices

- Keep your data clean and consistent.
- Use correct data types.
- Remove duplicates.
- Use parameters for dynamic data.
- Document your steps properly.
- Test the output on different datasets.
- Maintain source files and backups.

11. Useful DAX Operators

Operator	Description	Example
+	Addition	= Sales[Qty] + Sales[Price]
-	Subtraction	= Sales[Qty] - Sales[Cost]
*	Multiplication	= Sales[Qty] * Sales[Price]
/	Division	= Sales[Amount] / Sales[Qty]
=	Equal To	= Sales[Region] = "North"
<>	Not Equal To	= Sales[Region] <> "North"
>	Greater Than	= Sales[Amount] > 100

12. Sample DAX Measures

Measure	DAX Formula
Total Sales	= SUM(Sales[Amount])
Total Profit	= SUM(Sales[Profit])
Average Order Value	= AVERAGE(Sales[Amount])
Sales Growth	= DIVIDE([Total Sales] - [PY Sales], [PY Sales])
Total Customers	= DISTINCTCOUNT(Sales[CustomerID])



13. Conclusion

- DAX is powerful but easy to learn.
- It helps in creating accurate and dynamic reports.
- With practice, you can build advanced calculations and meaningful insights.



Master DAX, master insights!

The more you practice, the more powerful your reports become.



Topic-18

POWER BI MEASURES & COLUMNS (WHEN TO USE WHAT)

Date : _/_/___

Measures and Columns are both important in Power BI. It helps you create calculations, knowing when to use each one helps you build efficient and correct models.

Key Takeaway

- ★ Use column for storing data.
- Use measure for calculation.
- Both work together for better performance.



1. What are Columns?

- Columns store data in tables or in a model.
- Calculated columns are created during data refresh.
- They increase the size of the data model.
- Used for filtering, grouping, sorting and relationships.

Example: Columns in a Table

OrderID	Product	Quantity	Unit Price	Total Amount
1001	Laptop	2	500	1000
1002	Mouse	5	200	1000
1003	Keyboard	1	800	800
1004	Laptop	1	500	500
1005	Monitor	2	600	1200
...	...	...	...	...



Calculated Column Example:

Total = 'Sales'[Quantity] * 'Sales'[Unit Price]

2. What are Measures?

- Measures are calculated in the fly.
- They do not increase the size of the model.
- Used in values, KPIs and calculations.
- Usually work with filters and slicers.

Example: Measure Result

Product	Total Sales (Measure)
Laptop	1000
Mouse	1000
Keyboard	800
Grand Total	2,800



Measure Example:

Total Sales = SUM('Sales'[SalesAmount])

3. Key Differences

Aspect	Columns	Measures
Calculation Time	Calculated at data refresh time	Calculated at query time
Storage	Stored in the data model	Not stored (calculated on the fly)
Data Impact	Increases model size	Does not increase model size
Used For	Filtering, grouping, sorting, relationships	Values, KPIs, calculations
Changes with Filters	No, static	Yes, dynamic
Best For	Modeling data	Analysis and reporting

4. When to Use?

✓ Use Columns When:

- You need the value to be static.
- You need it to be used for grouping or filtering.
- You need it in relationships.
- You need it in slicers and sort.



✓ Use Measures When:

- You need dynamic calculations.
- You need aggregations (sum, avg, etc.).
- You are creating KPIs.
- You need it to work with slicers and filters.



5. Creating a Measure

- Go to Modeling tab.
- Click New Measure.
- Write the DAX formula.
- Press Enter.
- Use the measure in visuals.



Tip

- ✓ Rename measures (right click).
- ✓ Remove unnecessary steps.
- ✓ Keep transformation steps few.
- ✓ Use data types correctly.

Example:

Total Sales = SUM('Sales'[SalesAmount])

6. Common Scenarios

Scenario	Column	Measure
Use for slicers	Yes	Yes
Used for sorting	Yes	No
Used for filtering	Yes	Yes
Used in relationships	Yes	No
Dynamic calculation	No	Yes
Increases model size	Yes	No

7. Performance Impact

Columns (more impact on model size)

- Stored in memory.
- Increase file size.
- Can slow down refresh.



Measures (less impact)

- Calculated at runtime.
- No additional storage.
- Better for large models.



8. How to Create

- Go to Modeling tab.
- Click New Column or New Measure.
- Write the DAX formula.
- Press Enter.

New Column Example:

Age Group =
IF('Customers'[Age] >= 18,
"Adult", "Minor")

New Measure Example:

Total Sales =
SUM('Sales'[SalesAmount])

9. Useful DAX Operators

Operator	Description	Example
+	Addition	= Sales[Qty] + Sales[Price]
-	Subtraction	= Sales[Qty] - Sales[Cost]
*	Multiplication	= Sales[Qty] * Sales[Price]
/	Division	= Sales[Amount] / Sales[Qty]
=	Equal To	= Sales[Region] = "North"
<>	Not Equal To	= Sales[Region] <> "North"
>	Greater Than	= Sales[Amount] > 100

10. DAX Functions

Function	Description
SUM()	Adds up values
AVERAGE()	Calculates average
COUNT()	Counts non-blank values
CONCATENATE()	Joins text values
IF()	Returns value based on condition
TOTALYTD()	Calculates year-to-date totals
CALCULATE()	Changes filter context

11. Useful DAX Functions (More)

Function	Description
RELATED()	Get related values from another table
RELATEDTABLE()	Get all related rows from another table
LOOKUPVALUE()	Return a value from another table
ISBLANK()	Checks if value is blank
DISTINCTCOUNT()	Counts unique values

12. Conclusion

- ✓ DAX is powerful but easy to learn.
- ✓ Measures give flexibility and better performance.
- ✓ Columns help with modeling and relationships.
- ✓ Practice and real data build confidence.



13. Summary

- ✓ Use columns for data storage and modeling.
- ✓ Use measures for calculations and analysis.
- ✓ Choose the right one for better performance, accuracy and flexibility.
- ✓ Follow best practices for clean and efficient models.



Right Choice +
Good DAX =
Powerful Insights!



Build better models, make smarter decisions!



Topic-19

POWER BI DASHBOARDS (BRINGING IT ALL TOGETHER)

Date: ___/___/___

Dashboards in Power BI help you visualize important metrics, track KPIs and make data-driven decisions – anytime, anywhere.

Key Takeaway

- ★ Dashboards provide a single view of your key insights and help you monitor and achieve your goals. 😊

1. What is a Dashboard?

- A dashboard is a single page view of important metrics.
- It shows KPIs and key metrics in a glance.
- Dashboards are used for monitoring and tracking performance.



Goal

- ✓ Faster insights
- ✓ Better performance
- ✓ Informed decisions



Dashboards = At a glance
Reports = Deep dive

2. Dashboard vs Report

Feature	Dashboard	Report
Purpose	Monitor KPIs and key metrics	Explore and analyse data in detail
Layout	Single page	Multiple pages
Interactivity	High-level interactivity	Detailed drill and exploration
Best For	Quick insights and monitoring	In-depth analysis and storytelling

3. Dashboard Components

	Tiles	Visuals like cards, charts, KPIs and numbers.
	Slicers	Filters for interactive view (e.g., dropdowns).
	Visuals	Charts, graphs, tables, matrices, etc.
	Headers	Title, logo, navigation (brand elements).
	Buttons	Navigation or action buttons.

4. Creating a Dashboard

- 1 Go to Report view.
- 2 Click New Page.
- 3 Write the DAX formula.
- 4 Press Enter.
- 5 Use the measure in visuals.



Tip

- ✓ Rename visuals (right click).
- ✓ Remove unnecessary items.
- ✓ Keep transformation steps few.
- ✓ Use data types correctly.

Example:

Total Sales = SUM('Sales'[SalesAmount])

7. Performance Impact

- Columns (more impact on model size).
- Stored in memory.
- Increase file size.
- Can slow down refresh.



Measures (less impact)

- Calculated at runtime.
- No additional storage.
- Better for large models.



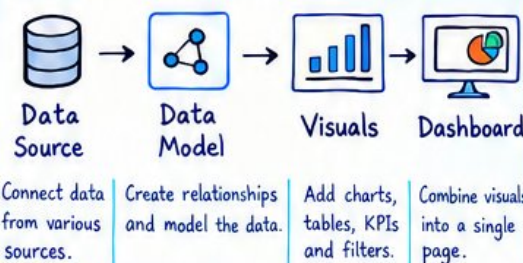
10. Useful DAX Operators

Operator	Description	Example
+	Addition	= Sales[Qty] + Sales[Price]
-	Subtraction	= Sales[Qty] - Sales[Cost]
*	Multiplication	= Sales[Qty] * Sales[Price]
/	Division	= Sales[Amount] / Sales[Qty]
=	Equal To	= Sales[Region] = "North"
<>	Not Equal To	= Sales[Region] <> "North"
>	Greater Than	= Sales[Amount] > 100

13. Summary

- ✓ DAX is powerful but easy to learn.
- ✓ Measures give flexibility and better performance.
- ✓ Columns help with modeling and relationships.
- ✓ Practice and real data build confidence.

5. Working with Visuals



Use visuals to tell a story with your data.

8. How to Create

- 1 Go to Modeling tab.
- 2 Click New Column or New Measure.
- 3 Write the DAX formula.
- 4 Press Enter.

New Column Example:

Age Group =
IF('Customers'[Age] >= 18,
"Adult", "Minor")

New Measure Example:

Total Sales =
SUM('Sales'[SalesAmount])

11. Useful DAX Functions

Function	Description
SUM()	Adds up values
AVERAGE()	Calculates average
COUNT()	Counts non-blank values
CONCATENATE()	Joins text values
IF()	Returns value based on condition
TOTALYTD()	Calculates year-to-date totals
CALCULATE()	Changes filter context

6. Common Scenarios

Scenario	Column	Measure
Use for slicers	Yes	Yes
Used for sorting	Yes	No
Used for filtering	Yes	Yes
Used in relationships	Yes	No
Dynamic calculation	No	Yes
Increases model size	Yes	No

9. Best Practices

- ✓ Keep your data clean and consistent.
- ✓ Use correct data types.
- ✓ Remove duplicates.
- ✓ Use parameters for dynamic data.
- ✓ Document your steps properly.
- ✓ Test the output on different datasets.
- ✓ Maintain source files and backups.



12. Sample DAX Measures

Measure	DAX Formula
Total Sales	= SUM(Sales[Amount])
Total Profit	= SUM(Sales[Profit])
Average Order Value	= AVERAGE(Sales[Amount])
Sales Growth	= DIVIDE([Total Sales] - [PY Sales], [PY Sales])
Total Customers	= DISTINCTCOUNT(Sales[CustomerID])

14. Quick Tips

- ✓ Test your DAX formulas.
- ✓ Use proper naming conventions.
- ✓ Keep your model simple.
- ✓ Iterate and improve.



Right Choice + Good DAX =
Powerful Insights!



A great dashboard turns data into action.

— Keep it simple, keep it focused, and keep improving! —

